

b) find the body's velocity, speed and acceleration at time $t = \frac{\pi}{4}$ (3)

Position $s(t) = \cos(t) - 3\sin(t)$ $s(\frac{\pi}{4}) = \frac{1}{\sqrt{2}} - \frac{3}{\sqrt{2}} = -\frac{2}{\sqrt{2}} = -\sqrt{2}$ $\frac{2}{\sqrt{2}} = \sqrt{2}$ m/s
 Velocity $s'(t) = -\sin(t) - 3\cos(t)$ $s'(\frac{\pi}{4}) = -\frac{1}{\sqrt{2}} - \frac{3}{\sqrt{2}} = -\frac{4}{\sqrt{2}} = -2\sqrt{2}$ $\frac{4}{\sqrt{2}} = 2\sqrt{2}$ m/s velocity
 Acceleration $s''(t) = -\cos(t) + 3\sin(t)$ $s''(\frac{\pi}{4}) = -\frac{1}{\sqrt{2}} + \frac{3}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$ $\frac{2}{\sqrt{2}} = \sqrt{2}$ m/s² = Acceleration

c) describe the motion of the body. (2)

Oscillates between many points and backwards sinusoidally.

Acceleration, and velocity, both oscillate as $s(t)$ oscillates.

Starts with speed of 3 m/s. Turns / changes direction around 5 and around

2 because the velocity change direction. Partition

1) 0/s at rest at those points.

Application (10 marks)

1. Let f be an increasing function with $f(0) = 2$. The derivative of f is given by $f'(x) = \sin(\pi x) + x^2 + 3$.

a) Write an equation for the line tangent to the graph of $g(x) = \frac{1}{f(x)}$ at $x=0$. (2)

at $x=0$ using chain rule
 $g'(x) = f'(x) \left(\frac{-1}{f(x)^2} \right)$ $y = -\frac{3}{4}(x) + 2\frac{1}{2}$ tangent
 $= f'(0) \left(\frac{-1}{f(0)^2} \right)$
 $g'(0) = -\frac{3}{4}$

b) Let g be the function defined by $h(x) = f(\sqrt{3x^2 + 4})$. Find $h'(2)$. (2)

$h'(x) = f'(\sqrt{3x^2 + 4}) \cdot \frac{d}{dx}(\sqrt{3x^2 + 4})$
 $= f'(4) \cdot \frac{1}{2\sqrt{3x^2 + 4}} \cdot (6x)$ $x=2$
 $= (\sin(4\pi) + 4^2 + 3) \left(\frac{1}{8} \right) \cdot (12)$
 $= (19)(1)(8)(2) = \frac{57}{2}$

$h'(2) = \frac{57}{2}$

2. The members of the Windfield society always divide the pavilion rental fee, for their occasions, equally among the members. Currently there are 65 members and the pavilion rents for \$250. The pavilion cost is increasing at a rate of \$10/per year, while the Windfield membership is increasing at a rate of 6 members/year. What is the current rate of change in each member's share of the pavilion rental fee? (3)

$S(x) = \frac{r(x)}{m(x)}$ $S(x) = \text{Share / member}$
 $m(x) = \text{members}$
 $r(x) = \text{Rental fee}$

$S'(x) = \frac{m(x)r'(x) - r(x)m'(x)}{m(x)^2}$

$= \frac{(65)(10) - (250)(6)}{(65)^2}$

$= \frac{-850}{65^2} = -0.20$
 $= -0.34 \$ / \text{member as the current rate of change}$

$\frac{2.5}{3}$

2. The cost, in dollars, of producing x units of a certain commodity is:

$$C(x) = 920 + 2x - 0.02x^2 + 0.00007x^3.$$

(3)

- a) Find the marginal cost function.

$$C'(x) = 2 - 0.04x + 0.00021x^2$$

- b) Find $C'(100)$ and explain its meaning.

$$C'(100) = 2 - 0.04(100) + 0.00021(100)^2$$

$$C'(100) = 21.1$$

This means is the marginal cost at 100, and it estimates the added cost for making the 101st item.

- c) Compare $C'(100)$ with the cost of producing the 101st item.

$$C(101) - C(100) = \text{cost of 101st item}$$

$$990.10107 - 990 = 0.10107 \text{ vs } 21.1$$

The values are quite close to each other.

Thinking (6 marks)

1. Determine $\frac{dy}{dx}$ if $y = \cos^2\left(\frac{1-2x}{1+2x}\right)$

$$y = \cos^2\left(\frac{1-2x}{1+2x}\right)$$

$$y' = \frac{d}{dx}\left(\cos\left(\frac{1-2x}{1+2x}\right)\right)^2 \cdot 2\left(\cos\left(\frac{1-2x}{1+2x}\right)\right)$$

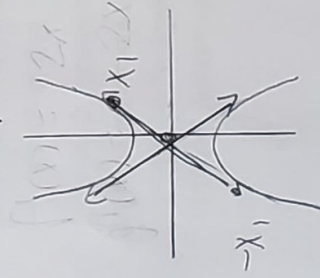
$$y' = \frac{d}{dx}\left(\frac{1-2x}{1+2x}\right) \cdot (-\sin\left(\frac{1-2x}{1+2x}\right)) \cdot 2\cos\left(\frac{1-2x}{1+2x}\right)$$

$$y' = \frac{-2}{(1+2x)^2} \cdot 2\left(\frac{1-2x}{1+2x}\right) \cdot (-\sin\left(\frac{1-2x}{1+2x}\right)) \cdot 2\cos\left(\frac{1-2x}{1+2x}\right)$$

2. How many lines are tangent to both $f(x) = 1 + x^2$ and $g(x) = -1 - x^2$? Find the coordinates of the points at which these tangents touch these curves.

(4) (3)

Both slopes must be equal for tangent to touch both



$$f'(x) = 2x$$

$$g'(x) = -2x$$

$$y = mx$$

$$y = f'(x_1)x$$

$$y = f(x_1)x$$

$$1 + x_1^2 = (2x_1)x$$

$$1 + x_1^2 = 2x_1^2$$

$$1 - x_1^2 = 0$$

$$\pm 1 = x_1$$

pairs of x that satisfy these 2 slopes are $(1, -1)$ $(2, -2)$... etc

$$\text{so } y = -2x$$

$$y = 2x$$

where else it intersects $f(x)$

for x_1 (the point of intersection)

Becomes $x_1 = -2x_1$

The points where these lines touch at

$$(1, 2) \quad (-1, 2)$$

$$(1, -2) \quad (-1, -2)$$

Knowledge/ Understanding (17 marks)

1. Differentiate the following functions.

(8)

a) $f(x) = \frac{x^{-7} - x^{-5}}{x^{-3}}$

$$f(x) = x^{-4} - x^{-2}$$
 Power rule

$$f'(x) = -4x^{-5} + 2x^{-3}$$

$$f'(x) = -4x^{-5} + 2x^{-3}$$

$$x \neq 0$$

b) $f(x) = (x^2 + 1)^3 \cdot (x^2 + 2)^6$
(write answer in factored form)

$$f'(x) = (x^2 + 1)^3 \frac{d}{dx} (x^2 + 2)^6 + (x^2 + 2)^6 \frac{d}{dx} [(x^2 + 1)^3]$$

$$= (x^2 + 1)^3 [2x(x^2 + 2)^5] + (x^2 + 2)^6 [6x(x^2 + 1)^2]$$

$$= 6x(x^2 + 2)^5(x^2 + 1)^2 [2(x^2 + 1) + (x^2 + 2)]$$

$$f'(x) = (6x)(x^2 + 2)^5(x^2 + 1)^2(3x^2 + 4)$$

c) $g(y) = \frac{(y-1)^4}{(y^2+2y)^5}$

(write answer in factored form)

$$\frac{dg}{dy} = \frac{(y^2+2y)^5 4(y-1)^3 - 5(y-1)^4 (2y+2)(y^2+2y)^4}{(y^2+2y)^{10}}$$

$$= \frac{(y-1)^3 [(y^2+2y) - 5(y-1)]}{(y^2+2y)^{10}}$$

$$= (y^2+2y)^4 (y-1)^3 (y-5y^2+2y)$$

$$= \frac{(y-1)^3 (y)(-5y^2+12)}{(y^2+2y)^6}$$

$$= \frac{-2(y-1)^3 (3y^2-4y-5)}{(y^2+2y)^6}$$

d) $y = \frac{\sec x}{1 + \tan x}$ (write answer in factored form)

$$y = \frac{1/\cos(x)}{1 + \sin(x)}$$

$$y = \frac{1/(\cos(x) \cos(x))}{\cos(x) + \sin(x)}$$

$$y = \frac{1}{\cos(x) (\cos(x) + \sin(x))}$$

such that $\cos(x) \neq 0$

$$y' = \frac{d}{dx} (\cos(x) + \sin(x))^{-1}$$

$$= \frac{\sin(x) - \cos(x)}{(\sin(x) + \cos(x))^2}$$

2. For questions a and c above, determine and explain values of x where function is not differentiable (2)

a) x is not differentiable at 0 due to a vertical asymptote at $x=0$, this makes $f(x)$ discont. at 0, meaning $f'(x)$ DNENot differentiable at $y = -2$ because it results division by 0, noting $g(y)$ not unit when $y = -2$.